

Low Power Design Using Double Edge Triggered Flip-Flops

Razak Hossain, Leszek D. Wronski, and Alexander Albicki

Abstract—In this paper we study the power savings possible using double edge triggered (DET), instead of, conventional single edge triggered (SET) flip-flops. We begin the paper by introducing a set of novel D-type double edge triggered flip-flops which can be implemented with fewer transistors than any previous design. The power dissipation in these flip-flops and single edge triggered flip-flops is compared via architectural level studies, analytical considerations and simulations. The analysis includes an implementation independent study on, the effect of input sequences, in the energy dissipation of single and double edge triggered flip-flops. The system level energy savings possible by using registers consisting of double edge triggered flip-flops, instead of single edge triggered flip-flops, is subsequently explored. The results are extremely encouraging, indicating that double edge triggered flip-flops are capable of significant energy savings, for only a small overhead in complexity.

I. INTRODUCTION

In recent years there has been an increasing demand for low power digital systems. Current trends indicate that architectural, logical and layout level techniques for reducing power dissipation will have to be incorporated in the integrated circuits of the future [1]. Double edge triggered (DET) flip-flops, which can maintain the same data rate at half the clocking frequency of conventional single edge triggered (SET) flip-flops, seem a natural design choice for building low power circuitry [2]–[5]. This is as DET flip-flops require the clock line to be charged and discharged half as often as SET flip-flops. Unfortunately, the applicability of DET flip-flops till now has been limited by the high overhead in complexity that they entail [3]. The increased complexity of the DET structures, which indicates greater energy consumption, negates some of the effects of a lower clock frequency. By defining a D-type DET flip-flop, however, in terms of its simplest functional representation we have designed a set of DET flip-flops which can be built with fewer transistors than any previously suggested implementation.

In Section II of this paper these flip-flops are introduced. In Section III we begin our study of energy dissipation by comparing the inherent tendency of SET and DET flip-flops to dissipate energy. Based on this approach, in Section IV, we provide a detailed analysis of the energy dissipation in the proposed DET flip-flops. In Section V we provide simulation results for the energy consumption and maximum data rates of the flip-flops. In Section VI we go beyond flip-flop level considerations to compare the energy consumption on a system level. Synchronous digital systems which have the same data rates, but which differ in terms of possessing SET and DET registers are studied. We end the paper in Section VII by providing some concluding remarks to our work.

Manuscript received February 11, 1993; revised July 29, 1993 and November 3, 1993.

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IEEE Log Number 9400600.

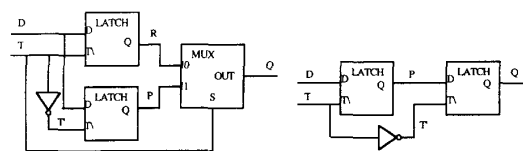


Fig. 1. Schematic diagram illustrating a SET and DET flip-flop.

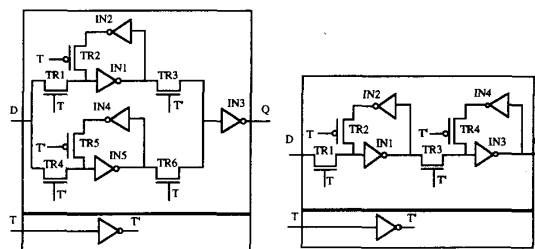


Fig. 2. Our implementation of a static SET and DET flip-flop.

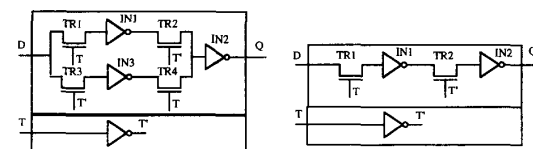


Fig. 3. Our implementation of a dynamic SET and DET flip-flop.

II. IMPLEMENTATION OF DOUBLE EDGE TRIGGERED FLIP-FLOPS

The design of our SET and DET flip-flop is based on the schematic diagrams shown in Fig. 1. Both the SET and DET flip-flops have two D-type latches. In the DET flip-flop the latches are arranged in parallel while in the SET flip-flop they are placed serially. Based on the schematics of Fig. 1, a set of CMOS SET and DET flip-flops have been designed. In Figs. 2 and 3, the static and dynamic version of these flip-flops are shown. Our design represents the simplest CMOS implementation of DET flip-flops. Despite their simplicity, the flip-flops have been shown by analytical and simulation means to be fully functional. In Tables I and II, we provide a comparison in terms of the total number of component transistors of our DET flip-flops, with those given in the literature [3]–[5]. Some of the columns in the tables contain two numbers separated by a hyphen. The integer on the left shows the number of transistors required in a flip-flop without a clock line inverter, while that on the right includes the two clock inverter transistors. Our flip-flops are seen to require the fewest transistors in both the static and dynamic category. As mentioned, reducing the complexity of double edge triggered flip-flops seems necessary before they are widely accepted. Early work in the field indicated a penalty in extra gates of between 50–100% for using DET flip-flops [2]. Our static and dynamic DET flip-flop on the other hand requires only four transistors more than a SET flip-flop.

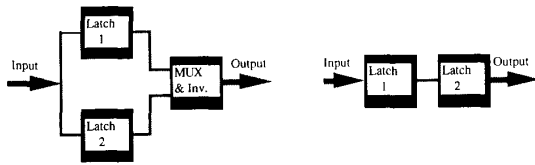


Fig. 4. Implementation independent model of a SET and DET flip-flop.

TABLE I
COMPARING STATIC DET FLIP-FLOP IMPLEMENTATIONS

Lu & Ercegovic [3]	Gagoet al. [5]	Implementation presented
26-28	16-18	16-18

TABLE II
COMPARING DYNAMIC DET FLIP-FLOP IMPLEMENTATION

Afghahi & Yuan [4]	Gago et al. [5]	Implementation presented
20	14-16	12-14

In the static flip-flops shown in Fig. 2, a p MOS transistor is used in the feedback path as it leads to a more compact layout than using a n MOS transistor. In a high noise or low voltage environment it may be necessary to replace the p MOS transistor with an n MOS device, or replace all the pass transistors with transmission gates. These requirements are not, however, investigated in this paper as our primary interest is to compare the system level power saving possible with DET flip-flops, for which purposes, the flip-flops used were fully functional.

III. ENERGY DISSIPATION AND INPUT SEQUENCES

In attempting to compare the putative advantages, in terms of power dissipation—of DET flip-flops over SET flip-flops—we are immediately forced to consider which input vectors are to be used for the purposes of comparison. This is as power dissipation in CMOS technology is a function not only of the supply voltage, circuit capacitances, data rates, and clocking frequency, but also, of the input vector set used. In order to illustrate the care that must be used in selecting an input sequence we shall use the simplified model for SET and DET flip-flops shown in Fig. 4. The following assumptions will be made about the energy dissipation in these structures:

- Energy dissipation occurs only in the latches and the inverting multiplexer when their states change. The latches and inverting multiplexer will be referred to as components in this section. The amount of energy dissipated in any component when it changes its logic state is the same. We are neglecting the effects of the clock line, since energy dissipation in them is independent of the input sequence.
- The inputs are applied as sequences of bits between periods of inactivity. Thus, inputs to the flip-flops before the sequence is applied, and after it has passed through, consist of logic zero bits.

The simplified nature of the model allows us to compare, in general implementation independent terms, the switching activity and hence energy dissipation in DET and SET flip-flops under different input vector sets. We shall illustrate our methodology with the following examples:

Example 1 Let us consider the input sequence 1010101000 and the changes it causes in the DET flip-flop. The input sequence entering latch 1 is 11110 which cause exactly two changes. In latch 2 the input sequence is 00000 which cause no changes. The entire input sequence enters the inverting multiplexer, causing eight changes. Thus, the total number of component changes is ten. In the SET flip-flop the input sequence causes eight changes in latches 1 and 2, or sixteen overall latch changes.

Example 2 As our second input vector set let us consider the following sequence 1100110000. In the DET flip-flop this input vector set causes twelve total component changes. In the SET flip-flop the same sequence, however, causes eight component changes.

These two examples suggest that input vector sets of the form 10101010 etc. cause more changes in a SET flip-flop, while input sequences of the form 11001100 etc. induce a greater number of changes in DET system. What input vector should hence be used in simulation work, i.e., what finite sequence of bits will best approximate the switching activity of a random sequence of the same length? This requires us to know the expected number of component changes in a SET and DET flip-flop for a random sequence of input bits. In Lemma 1 we deal with this issue.

Lemma 1: An input vector set consisting of n bits—equally likely of being one or zero and independently distributed, and which are applied to a flip-flop between idle periods (input at logic state zero)—will have the ratio of expected component changes in a double edge triggered to a single edge triggered flip-flop of $(2n+3)/(2n+2)$.

Sketch of the Proof The n bit sequence causes $(n+1)/2$ expected changes in latches 1 and 2 of the SET flip-flop. This sums to $(n+1)$ overall expected component changes. In the DET flip-flop, the n bits cause $(n+1)/2$ expected changes in the inverting multiplexer, and $(n/2+1)/2$ expected changes in latches 1 and 2. This gives a total of $(2n+3)/2$ expected changes. The ratio of expected changes in the DET to SET flip-flop is, thus, $(2n+3)/(2n+2)$.

Lemma 1 indicates that for finite input sequences, applied between periods of no activity, the expected number of component changes in the data path of a DET flip-flop is greater than in a SET flip-flop. For values of $n \geq 10$, however, this difference becomes less than 5%.

IV. ANALYTICAL COMPARISONS OF ENERGY DISSIPATION IN FLIP-FLOPS

A more detailed analysis of the expected energy dissipation is possible when specific implementations are considered. We shall assume, as in Section III, that a sequence of n bits—equally likely of being one and zero and independently distributed—is applied between idle periods, i.e., logic state zero. The drain regions are also assumed not to be shared between any transistors. We shall calculate the expected capacitive switching in the flip-flops by summing the expected switching capacitance across all k nodes of the system:

Expected switching capacitance

$$= \sum_{i=1}^k (\text{Capacitance of node } i) \times (\text{Number of transitions at node } i)$$

Let us analyze the dynamic SET flip-flop given in Fig. 3. We can divide the switching capacitance of the circuit into two components: the capacitance associated with the data path, and that associated with the clocking circuitry. The data path capacitance consists of the drain capacitances (c_d 's) of the transistors TR1 and TR2, as well as, the drain and gate capacitances (c_g 's) associated with the

inverters IN1 and IN2. The sum of the data path capacitance is $8c_d + 4c_g$. The capacitance of the clocking circuitry consists of the gate capacitance contributed by the transistors TR1 and TR2 the drain and gate capacitance of the clock line inverter, and, c_{local} , the local inter-connect capacitance for the clock line between two adjacent flip-flops in a register. This comes to $2c_d + 4c_g + c_{local}$. It can be easily shown that for a sequence of n bits, the data path capacitance will have an expected switching of $(n+1)/2$, while requiring $(n+1)$ clock pulses or $2(n+1)$ clock transitions to propagate through the flip-flop and return it to its initial condition. Hence, the expected capacitive switching in the dynamic SET flip-flop is: $(8c_d + 4c_g)(n+1)/2 + (2c_d + 4c_g + c_{local})2(n+1) = (10c_d + 8c_g + 2c_{local})(n+1)$.

The calculation of the expected capacitive switching for the dynamic DET is complicated by the fact that two parallel data paths exist in the flip-flop. All n bits enter and exit the flip-flop through the same path, but half the data travels through the inverter IN1, while the other half traverses through IN2. The path that is common for all n bits contains the drain capacitances of the transistors TR1 and TR3 facing the input D , the drain capacitance of TR2 and TR4 facing the output, and, the inverter IN2. This has a total capacitance of $6c_d + 2c_g$ with an expected switching activity of $(n+1)/2$. The top and bottom path each contain two drain capacitances and an inverter, i.e., $4c_d + 2c_g$, with an expected switching activity of $(n/2 + 1)/2$. The clocking circuitry has a capacitance of $2c_d + 6c_g + c_{local}$ and requires $(n+2)$ clock transitions to clock the sequence and return the flip-flop to its initial state. Thus, the total expected capacitive switching in the dynamic double edge triggered flip-flop is: $(6c_d + 2c_g)(n+1)/2 + 2 \times (4c_d + 2c_g)(n/2 + 1)/2 + (2c_d + 6c_g + c_{local})(n+2) = (7c_d + 8c_g + c_{local})(n) + (11c_d + 15c_g + 2c_{local})$.

If we assume that the only energy dissipation in the flip-flops occurs when the node capacitances are charged and discharged, then the value of the energy dissipated can be calculated by using the well known formula $(0.5)CV^2$, where C is the switching capacitance and V is the supply voltage. We shall assume that n is very large, which is equivalent to continuous data flow, so that all terms which do not contain n can be neglected. The ratio of energy dissipated in the dynamic DET to SET flip-flop is then: $\frac{7c_d + 8c_g + c_{local}}{(8c_d + 10c_g + 2c_{local})}$, which is always less than one.

Example 3 We can calculate the ratio of energy dissipated in a dynamic SET and DET flip-flop under continuous data flow using MOSIS 2.0 μm CMOS SPICE data. The average value of c_g , for $p\text{MOS}$ and $n\text{MOS}$ transistors, is 8.35 fF. The value of c_d is 15.35 fF. We assume that the metal inter-connect between adjacent flip-flops is $20 \mu\text{m} \times 5 \mu\text{m}$. This gives a c_{local} value of 8.8 fF. The ratio of energy dissipated in the DET to SET flip-flop is: $[7(15.35) + 8(8.35) + 8.8] / [8(15.35) + 10(8.35) + 2(8.8)] = 0.82$. When no data flows the ratio of energy dissipated in the DET to SET flip-flop is 0.61.

Using methods similar to that used for the dynamic structures, the steady state ratio of energy dissipated in a static DET to SET flip-flop is: $\frac{9c_d + 11c_g + c_{local}}{12c_d + 16c_g + 2c_{local}}$, which is also always less than one. This ratio is 0.71 for the capacitive values given in Example 3. When no data flows, the ratio of energy dissipated in a DET flip-flop to that in a SET flip-flop reduces to 0.59. The analytical results, thus suggest, that DET flip-flops consume significantly less energy than SET flip-flops.

V. SIMULATION RESULTS FOR FLIP-FLOPS

While our analytical results were detailed, they could not of course take into account all of the factors that affect the energy dissipation in a flip-flop. These factors include short circuit and leakage currents,

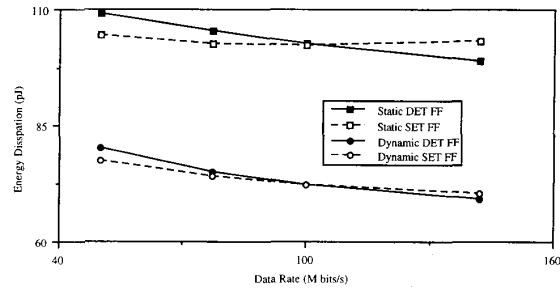


Fig. 5. Energy dissipation in the static and dynamic, SET and DET flip-flops.

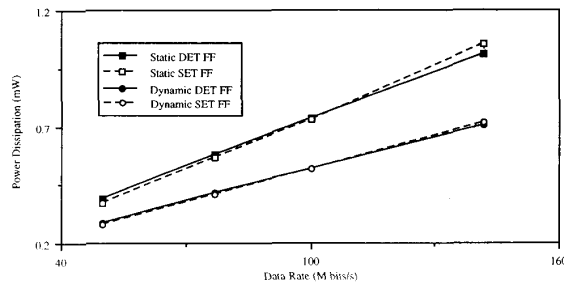


Fig. 6. Power dissipation in the static and dynamic, SET and DET flip-flops.

the partial (as opposed to complete) charging and discharging of nodes, and, the effects of other parasitic capacitances. To consider these factors, a set of static and dynamic, double and single edge triggered flip-flops were laid out. The flip-flops were then simulated for 2.0 μm CMOS technology to determine their energy dissipation and maximum data rate. The energy dissipation was measured using Fisher's method [6], for the input sequence 10101100110000. This sequence causes exactly 16 component changes in a DET and SET flip-flop, resulting in a ratio of component changes of 1. From Lemma 1, in Section III, we know that for a random sequence of length $n = 14$ the ratio of component changes in a DET to SET flip-flop is 1.033. Thus, as the sequence chosen had switching characteristics similar to a random sequence of the same length, it was suitable for measuring energy consumption. Data rates of 50 to 142 Mbits/s were used during energy measurement. The results are shown in Figs. 5 and 6.

Fig. 5 shows that the energy dissipation in the dynamic structures is about 30% less than in the static structures. The energy and power consumption in SET and DET flip-flops are approximately equivalent. Figs. 5 and 6 also show that while DET flip-flops dissipate less energy than SET flip-flops at high frequencies, this is not true at lower frequencies. The discrepancy with the analytical results may be attributed to the fact that at lower frequencies capacitors tend to charge and discharge more completely than at higher frequencies, causing greater energy consumption. Since the clocking circuitry in the DET flip-flop is larger, and consequently has a greater capacitive load, it is more seriously affected. The deviation from analytical results was exacerbated by the implementation of the flip-flops with a single metal layer, which significantly increased, c_{local} , and caused a discrepancy in its value between the SET and DET flip-flop.

Fig. 5 at first glance seems to contradict the popular belief that lowering "the clock and data transmission rates" will result in

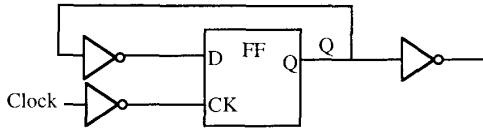


Fig. 7. Simulation set-up used to measure maximum data rates in flip-flops.

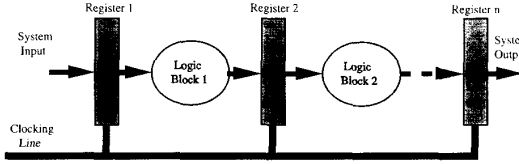


Fig. 8. Model used for comparing system level energy dissipation.

lower energy consumption [7]. The paradox can be explained by remembering that our measurement of energy dissipation is based on a fixed amount of data which has to be processed. If we had merely been measuring the average energy dissipation in two identical systems operating at different frequencies, under continuous data flow, then the system operating at a lower frequency would obviously dissipate less energy. In our simulations we also measured the energy dissipated in single and double edge triggered flip-flop under idle input conditions. With data rates of 100 Mbits/s, and inputs set at zero, the ratio of energy consumption in a static DET to SET flip-flop was measured at $(35.5 \text{ pJ})/(52.2 \text{ pJ}) = 0.68$. The corresponding ratio for dynamic flip-flops is $(26.6 \text{ pJ})/(40.1 \text{ pJ}) = 0.66$. This data allows us to consider how idle inputs affect energy dissipation.

Example 4 Let us compare the energy dissipated in a SET and DET flip-flop assuming that new data enters the flip-flops 20% of the time. Since data flows only intermittently in most applications this assumption is reasonable. If the data rate is 100 M bits/s then the energy consumption in a static DET to SET flip-flop, using the data measured in the simulations, is $[(102.7 \text{ pJ})(0.2) + (35.5 \text{ pJ})(0.8)]/[(102.3 \text{ pJ})(0.2) + (52.2 \text{ pJ})(0.8)] = (48.94 \text{ pJ})/(62.22 \text{ pJ}) = 0.79$. The corresponding value for a dynamic DET flip-flop is $[(72.2 \text{ pJ})(0.2) + (35.5 \text{ pJ})(0.8)]/[(72.4 \text{ pJ})(0.2) + (52.2 \text{ pJ})(0.8)] = (42.84 \text{ pJ})/(56.24 \text{ pJ}) = 0.76$.

In addition to measuring the energy consumption of the flip-flops, we also studied the maximum data rate possible in these structures. This was done by the circuit shown in Fig. 7. The output, Q , of the flip-flop was observed to note the maximum frequency for which the input, D , was properly sampled. The maximum data rate for static SET and DET flip-flops was 125 Mbits/s. The maximum data rate for the dynamic SET and DET flip-flop was 167 and 154 Mbits/s. While the dynamic SET flip-flop is capable of higher data rates, one must remember that it requires a clock rate twice that of a DET flip-flop.

VI. SYSTEM LEVEL COMPARISONS OF ENERGY CONSUMPTION

The analysis and simulation work till now has concentrated on the flip-flop level energy savings possible through the use of double edge triggered flip-flops. While this is important, the overall system level energy savings possible via the use of double edge triggered flip-flops is more so. We have formulated a model, shown in Fig. 8, to compare system level energy dissipation when DET and SET flip-flops are used in registers.

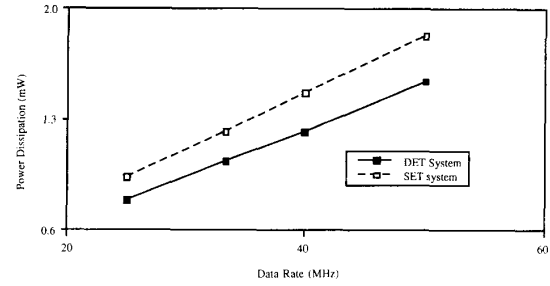


Fig. 9. Power dissipation for the circuit in Example 5.

Fig. 8 shows a combinational circuit consisting of $n - 1$ logic blocks and n registers. Energy is dissipated in the logic blocks, the registers, and the clocking line. We will study the energy dissipated in each of these subsystems for a single time period between successive inputs entering the system. Let the expected energy dissipation in the combinational logic circuitry and the clock line be E_{logic} and E_{clock} . If E_{set} and E_{det} is the energy dissipated in each SET and DET flip-flop, and each register has m flip-flops, then the energy dissipated in the n registers is $(mn)E_{\text{set}}$ for SET registers, and, $(mn)E_{\text{det}}$ for DET registers.

Thus, for the SET flip-flop system the energy dissipation is: $E_{\text{logic}} + 2E_{\text{clock}} + (mn)E_{\text{set}}$, while for a DET flip-flop system it is: $E_{\text{logic}} + E_{\text{clock}} + (mn)E_{\text{det}}$. Since the SET flip-flops require two clock transitions during each time period, as opposed to one clock transition for a DET flip-flop, the coefficient for E_{clock} is two in the SET register system. From Section IV we know that $E_{\text{set}} > E_{\text{det}}$. This leads to the obvious conclusion that the DET system dissipates less energy as: $E_{\text{logic}} + E_{\text{clock}} + (mn)E_{\text{det}} < E_{\text{logic}} + 2E_{\text{clock}} + (mn)E_{\text{set}}$. In order to quantify the extra energy dissipated in the SET system, we subtract the energy dissipated in the DET system from that in the SET system. This gives us the following relation: the extra energy consumption in the SET register system during one clock period = $E_{\text{clock}} + (mn)(E_{\text{set}} - E_{\text{det}})$.

The relation leads to the following observations. First, the difference in energy dissipated increases with the extent of the clocking circuitry (E_{clock}) and the degree of pipelining (mn). Second, as the difference in energy consumption is in a single period, it increases linearly with clock frequency. Since increasing clock frequencies and pipelining are a trend in the field of VLSI, the possibility for increased energy savings through the use of DET flip-flops exists. Our analysis has not, however, provided us with an overall sense of the magnitude of energy savings possible with DET flip-flops. In Example 5 we study this issue for a particular circuit.

Example 5 The logic circuit used was a 5 bit ripple carry adder. The effect of the global clock line was simulated with a clock line of length $300 \mu\text{m}$, clocked via a buffer circuit. Two designs were produced. One of them had a static 6-bit SET register at the output, while the other had a static 6-bit DET register at the output. The energy dissipated was measured for a set of inputs with data rates varying from 25 to 50 Mbits/s. The input vectors used had two changing inputs followed by four idle clocking periods. The area of the adder and SET register was $579 \mu\text{m} \times 250 \mu\text{m}$, while the area for the DET system was $595 \mu\text{m} \times 286 \mu\text{m}$. This corresponds to a 17% overhead for the DET design. In most actual circuits, however, more combinational logic would be present between registers, significantly reducing the area overhead. The results of the simulation are given in Fig. 9.

In Fig. 9 the power dissipation of the double and single edge triggered systems are shown. It can be seen that the single edge triggered system has a higher level of power dissipation. For example, at 33 Mbits/s the SET system has a power dissipation equal to the DET system operating at 40 Mbits/s. Furthermore, under idle conditions when no new data enters the system, the ratio of energy dissipated in DET to SET systems is 0.62. The actual energy dissipation in the SET system was about 130 pJ higher than in the DET system, which translates to a 17% greater energy dissipation in the SET system.

VII. CONCLUSION

DET flip-flops have been shown to have lower energy requirements ($\sim 20\%$) than SET flip-flops, with only a limited overhead in complexity. This has been shown by analytical and simulation means. DET and SET flip-flops have comparable maximum data rates—the dynamic SET flip-flops being slightly faster, but requiring a clock operating at twice the frequency as a dynamic DET flip-flop. Simulation results also indicated that dynamic flip-flops consume less energy than static flip-flops. In addition to the energy savings possible in the devices themselves, system level energy savings are also possible with the use of DET flip-flops. A conservative example provides encouraging results in terms of the energy savings ($\sim 17\%$) possible via DET flip-flops.

ACKNOWLEDGMENT

The authors wish to thank the reviewers of the paper for their many helpful comments.

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Algorithms and Bounds for Layer Assignment of MCM Routing

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Abstract—We present new algorithms for the layer assignment problem of *Multichip modules* (MCM's). Our algorithms produce results that require between 70% and 25% of the number of layers required by the previous algorithms. We also present a new model for the problem that results in a better utilization of the routing area of the MCM, thus reducing the number of required layers even more. We provide lower and upper bounds on the performance of our algorithms which are tighter than the ones obtained before. Through our experimental results we show that the solutions obtained by our algorithms are close to the lower bounds.

I. INTRODUCTION

The problem of *layer assignment* for VLSI and multichip modules (MCM) has received significant attention recently [3], [4], [9], [12], [14], [17]–[20]. In this paper we present new algorithms for the layer assignment problem of MCM's and show that the number of layers required by the solutions obtained by our algorithms require between 70% and 25% of the number of layers required by the previous algorithms [12]. Additionally, our experimental results indicate that the solutions obtained by our algorithms are close to the lower bounds.

MCM's are compact high performance packages that interconnect several, tens or hundreds, of chips. MCM's allow many bare integrated circuit dies to be mounted onto the top layer, called *Chip Layer*, of a multilayer laminated, ceramic or thin film module, called *Signal Distribution Layer*, which provides metalized wiring layers for signal, power, and ground connections. Elimination of individual chip packages can significantly reduce *inter-chip propagation delay*, *size*, *weight*, *cost*, and *power consumption* of the system. Connecting the IC's in closer proximity will enable systems to run at a higher *clock frequency*. Since inter-chip connections are done within the MCM, these modules are more durable and can withstand much harsher environments than *Printed Circuit Boards*. MCM's consisting of up to 63 layers have been manufactured for various applications and at reasonably low cost, [1], [2], [5], [8], [10], [16]. Fig. 1 depicts the essential elements of an MCM as defined in [12], where chips are placed on the *Chip Layer*, *Pin Redistribution Layer* distributes pins uniformly, and *Signal Distribution Layer* provides the required connections of the pins.

A central problem to the design of MCM's is the layer assignment problem, which determines the number of layers to be used for a feasible routing of the nets. Since the number of layers required is directly related to the size, cost, and the power requirement of the final product, minimizing the number of layers is very important. With more layers the module becomes thicker (height of the module increases), more power is required, and more heat is generated, hence, more heat removing and heat dissipating devices are

Manuscript received May 6, 1993; revised November 8, 1993.

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IEEE Log Number 9400601.